



WORK SHEET - 21

INDEFINITE & DEFINITE INTEGRATION

Q.1 (A)

$$f(x) = \int_0^x (1+t^3)^{-1/2} dt$$

$$\Rightarrow f[g(x)] = \int_0^{g(x)} (1+t^3)^{-1/2} dt$$

$$\Rightarrow x = \int_0^{g(x)} (1+t^3)^{-1/2} dt$$

[$\because$ g is inverse of $f \Rightarrow f[g(x)] = x$]

Differentiating with respect to x , we have

$$1 = (1+g^3(x))^{-1/2} \cdot g'(x) \Rightarrow (g'(x))^2 = 1+g(x)^3$$

Differentiating again with respect to x , we have

$$2g'(x)g''(x) = 3(g(x))^2 \cdot g'(x) \Rightarrow \frac{g''(x)}{g^2(x)} = \frac{3}{2} \text{ Ans.}$$

Q.2 (A)

$$\therefore \int (x^9 + x^6 + x^3)(2x^6 + 3x^3 + 6)^{\frac{1}{3}} dx$$

$$= \int (x^8 + x^5 + x^2)(2x^9 + 3x^6 + 6x^3)^{\frac{1}{3}} dx$$

$$\text{Let } 2x^9 + 3x^6 + 6x^3 = t$$

$$\Rightarrow 18(x^8 + x^5 + x^2)dx = dt$$

$$\therefore I = \int \frac{t^{1/3}}{18} dt = \frac{1}{18} \cdot \frac{t^{4/3}}{4/3} + C = \frac{1}{24} t^{4/3} + C$$

$$\therefore AB = 24 \times \frac{4}{3} = 32 \text{ Ans.}$$

Q.3 (C)

$$\int e^x (\tan x - x + \tan^2 x - \tan^2 x - 2 \tan x \sec^2 x) dx$$

$$= \int e^x (\tan x - x + \tan^2 x) dx - \int e^x (\tan^2 x + 2 \tan x \sec^2 x) dx$$

$$= e^x (\tan x - x - \tan^2 x) + C$$

$$f(x) = \tan x - x - \tan^2 x$$

$$f\left(\frac{\pi}{4}\right) = \frac{-\pi}{4}.$$



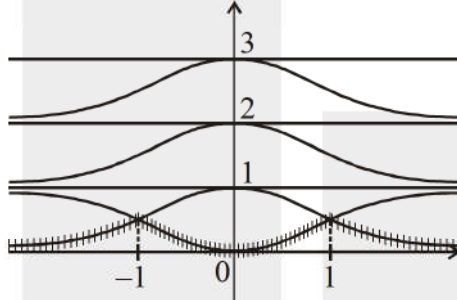
Q.4 (B)

$$f(x) = \left| [\sin x - 2\sqrt{2} \cos x + 2] + \frac{1}{1+t^2} \right|, x, t \in \mathbb{R}$$

$$-1 \leq \sin x - 2\sqrt{2} \cos x + 2 \leq 5$$

$$\therefore [\sin x - 2\sqrt{2} \cos x + 2] = -1, 0, 1, 2, 3, 4, 5$$

$$g(t) = \text{Min. } (f(x)) = \text{Min. } \left(\left| -1 + \frac{1}{1+t^2} \right|, \frac{1}{1+t^2}, 1 + \frac{1}{1+t^2}, \dots, 5 + \frac{1}{1+t^2} \right)$$



$$\therefore \int_{-\infty}^{\infty} g(t) dt = 2 \int_0^{\infty} g(t) dt = 2 \left(\int_0^1 \left(1 - \frac{1}{1+t^2} \right) dt + \int_1^{\infty} \frac{1}{1+t^2} dt \right) = 2. \text{ Ans.}$$

Q.5 (A)

$f(x) = (\cot x + \cot 2x + \dots + \cot 10x)$ is periodic with period π and $f(\pi - x) = -f(x)$

$$\Rightarrow \int_0^{100\pi} f(x) dx = 100 \int_0^{\pi} f(x) dx = 100 \int_0^{\pi} f(\pi - x) dx = -100 \int_0^{\pi} f(x) dx \Rightarrow 0 \text{ Ans.}$$

Q.6 (C)

$$\int_1^3 f(x) dx = \int_1^2 x dx + \int_2^3 2(x-1) dx = \left. \frac{x^2}{2} \right|_1^2 + \left. (x^2 - 2x) \right|_2^3$$

$$= \frac{1}{2} (4 - 1) + (9 - 6) - (4 - 4)$$

$$= \frac{3}{2} + 3 = \frac{9}{2} \text{ Ans.}$$

Q.7 (C)

$$\text{Put } x = \sqrt{2} \sin \theta \Rightarrow I = \sqrt{2} \int \frac{\sin \theta d\theta}{1 + \sqrt{2} \cos \theta} = -\ln |1 + \sqrt{2} \cos \theta| + c$$

$$\therefore I = -\ln |1 + \sqrt{2 - x^2}| + c \text{ Ans.}$$

Q.8 (C)

Differentiate both sides with respect to x, we get comparing

$$A = \frac{-1}{16} \text{ Ans.}$$

Q.9 (B)

$$\text{Let } I = \int_{-3}^3 x^8 \{x^{11}\} dx \quad \dots\dots(i)$$

$$\text{using king, } I = \int_{-3}^3 x^8 \{-x^{11}\} dx \quad \dots\dots(ii)$$

using (i) + (ii), we get

$$2I = \int_{-3}^3 x^8 dx = 2 \int_0^3 x^8 dx$$

$$I = \left(\frac{x^9}{9} \right)^3 = 3^7 \text{ Ans.}$$

Q.10 (B)

$$P_n = \int_1^e (\ln x)^n \cdot \frac{1}{x} dx$$

$$\Rightarrow P_n = e - n P_{n-1}$$

Q.11 (B)

Q.12 (C)

Q.13 (C)

$$f(x) = 12x^2 \int_0^1 y f(y) dy + 20x \int_0^1 y^2 f(y) dy + 4x$$

$$f(x) = x^2 A + xB + 4x \quad \text{i.e. a quadratic polynomial}$$

$$\Rightarrow f(x) = 12x^2 \int_0^1 y(Ay^2 + By + 4y) dy + 20x \int_0^1 y^2(Ay^2 + By + 4y) dy + 4x$$

$$x^2 A + xB + 4x = 12x^2 \left[\frac{A}{4} + \frac{B}{3} + \frac{4}{3} \right] + 20x \left[\frac{A}{5} + \frac{B}{4} + \frac{4}{4} \right] + 4x$$

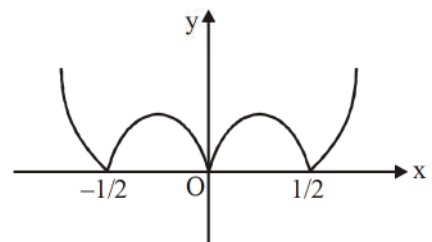
$$\Rightarrow A = 3A + 4B + 16 ; \quad B = 4A + 5B + 20$$

$$A = -2, \quad B = -3$$

$$\Rightarrow f(x) = -2x^2 + x$$

$$f_{\max} = \frac{1}{8} \text{ at } x = \frac{1}{2}$$

$$\text{and } \int_0^1 [f(x)] dx = \int_0^{1/2} 0 + \int_{1/2}^1 -1 dx = \frac{-1}{2} \text{ Ans.}$$



Q.14 (AB)

$$I_{m,n} = \int_0^1 x^{n-1} (\log x)^m dx$$

$$= (\log x)^m \cdot \frac{x^n}{n} \Big|_0^1 - \frac{m}{n} \int_0^1 (\log x)^{m-1} \cdot \frac{1}{x} \cdot x^n dx$$

$$I_{m,n} = 0 - \frac{m}{n} I_{m-1,n}$$

$$k = \frac{I_{m,n}}{I_{m-1,n}} = -\frac{m}{n} \Rightarrow \frac{I_{r,52}}{I_{r-1,52}} = \frac{-r}{52}$$

$$\Sigma k = -\frac{1}{52} (1 + 2 + \dots + 51) = \frac{-52 \times 51}{2} = \frac{-51}{2} \text{ Ans.}$$

Q.15 (BC)

Let degree of $f(x)$ be n .

$\therefore$ degree of $f(f(x))$ will be n^2

$$\therefore n^2 = n \Rightarrow n = 1$$

$\therefore f(x)$ will be a linear polynomial

$$\text{Let } f(x) = ax + b$$

$$\therefore f(f(x)) = 12x + 10 - f(x)$$

$$\Rightarrow a(ax + b) + b = 12x + 10 - (ax + b)$$

$$\therefore a^2 = 12 - a \Rightarrow a^2 + a - 12 = 0 \Rightarrow (a + 4)(a - 3) = 0 \Rightarrow a = -4 \text{ or } 3$$

$$\text{and } ab + b = 10 - b$$

(i) If $a = -4$, then $-4b + b = 10 - b \Rightarrow -2b = 10 \Rightarrow b = -5$

$$\therefore f(x) = -4x - 5$$

$$\Rightarrow \int_{-1}^1 f(x) dx = -10$$

(ii) If $a = 3$, then $3b + b = 10 - b \Rightarrow 5b = 10 \Rightarrow b = 2$

$$\therefore f(x) = 3x + 2 \Rightarrow \int_{-1}^1 f(x) dx = 4$$

Q.16 (ABC)

$$\therefore \frac{f(x)}{x(x+1)^2(x-2)^3} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{(x+1)^2} + \frac{D}{x-2} + \frac{E}{(x-2)^2} + \frac{F}{(x-2)^3}$$

$$\int \frac{f(x)}{x(x+1)(x-2)^3} dx \text{ will be a logarithmic function if } C = E = F = 0$$

$$\therefore f(x) = (x+1)(x-2)^2$$

$$f(1) = 2, f'(2) = 0$$



$$\int_0^4 f(x) dx = \int_0^4 (x+1)(x-2)^2 dx$$

Putting $x-2=t$

$$= \int_{-2}^2 (t+3)t^2 dt = 2(t^3)_0^2 = 16 \text{ Ans.}$$

Q.17 (ABCD)

$$I = \int \frac{\tan x dx}{5+3 \tan^2 x} = \int \frac{\sin x \cos x dx}{5 \cos^2 x + 3 \sin^2 x}$$

$$\text{Let } 5 \cos^2 x + 3 \sin^2 x = t$$

$$-10 \cos x \sin x + 6 \sin x \cos x = \frac{dt}{dx}$$

$$\sin x \cos x dx = \frac{-dt}{4}$$

$$\Rightarrow I = \int \frac{-dt/4}{t} = -\frac{1}{4} \log |t| + c = -\frac{1}{4} \log |5 \cos^2 x + 3 \sin^2 x| + c \text{ Ans.}$$

Q.18 [0000]

$$A = \int_1^{\tan \theta} \frac{t dt}{1+t^2}, \text{ Let } t = \frac{1}{z} \Rightarrow dt = \frac{-1}{z^2} dz$$

$$A = - \int_1^{\cot \theta} \frac{dz}{z(z^2+1)} = -B \Rightarrow A+B=0$$

In determinate Column-1 and Column-2 will be identical

Q.19 [5]

$$f(x) = \lambda e^{-x}$$

$$f'(x) = -\lambda e^{-x}$$

$$f'(0) - f'(0) = \lambda - (-\lambda) = 2\lambda = \pi / \ln 5$$

$$\text{Now, } \lambda = \int_{1/a}^a \frac{\tan^{-1} t}{t} dt \dots\dots(i)$$

$$\text{Putting, } t = \frac{1}{y} \Rightarrow dt = \frac{-1}{y^2} dy$$

$$\lambda = \int_a^{1/a} \frac{\tan^{-1} \frac{1}{y}}{\frac{1}{y}} \left(\frac{-1}{y^2} \right) dy$$

$$\lambda = \int_{1/a}^a \frac{\tan^{-1} \frac{1}{y}}{y} dy \dots\dots(ii)$$

$$2\lambda = \int_{1/a}^a \frac{\left(\frac{\pi}{2}\right)}{y} dy$$

$$2\lambda = \frac{\pi}{2} (\ln y)_{1/a}^a = \frac{\pi}{2} \cdot 2 \ln a = \pi \ln a$$

$$\therefore \pi \ln a = \pi \ln 5 \Rightarrow a = 5$$

Q.20 [3]

Put $e^x = t$

$$I_n = \int \frac{t^n dt}{\left(1 + t + \frac{t^2}{2!} + \dots + \frac{t^n}{n!}\right)} = n! \int \frac{1 + t + \frac{t^2}{2!} + \dots + \frac{t^n}{n!} - \left(1 + t + \frac{t^2}{2!} + \dots + \frac{t^{n-1}}{(n-1)!}\right)}{\left(1 + t + \frac{t^2}{2!} + \dots + \frac{t^n}{n!}\right)} dt$$

$$= n! \left(t - \int \frac{1 + t + \frac{t^2}{2!} + \dots + \frac{t^{n-1}}{(n-1)!}}{1 + t + \frac{t^2}{2!} + \dots + \frac{t^n}{n!}} dt \right)$$

$$\text{Let } 1 + t + \frac{t^2}{2!} + \dots + \frac{t^n}{n!} = v ; \quad dv = 1 + t + \frac{t^2}{2!} + \dots + \frac{t^{n-1}}{(n-1)!} dt$$

$$\therefore I_n = n! \left(t - \ln \left(1 + t + \frac{t^2}{2!} + \dots + \frac{t^n}{n!} \right) \right) + c$$

$$= n! \left(e^x - \ln \left(1 + e^x + \frac{e^{2x}}{2!} + \dots + \frac{e^{nx}}{n!} \right) \right) + c$$

$$\therefore g(x) = \lim_{n \rightarrow \infty} \ln \left(1 + e^x + \frac{e^{2x}}{2!} + \dots + \frac{e^{nx}}{n!} \right) = \ln(e^{e^x}) = e^x$$

$\therefore$ Number of solutions of $e^x = x^2$ is 3. **Ans.**

Q.21 [0200]

$$1 - \frac{\sin^2 x}{1 + \cos x} + \frac{1 + \cos x}{\sin x} - \frac{\sin x}{1 - \cos x}$$

$$= 1 - (1 - \cos x) + \frac{1 + \cos x}{\sin x} - \frac{\sin x}{1 - \cos x} = \cos x + \left[\frac{1 - \cos^2 x - \sin^2 x}{\sin x(1 - \cos x)} \right] = \cos x$$

$$\int_0^{100\pi} |\cos x| dx = 100 \left(\int_0^{\pi/2} \cos x dx + \int_0^{\pi} -\cos x dx \right) = 200 \text{ Ans.}$$

